CORE PURE / PROOF BY INDUCTION

Proof by Induction - Divisibility

Questions by topic 8 questions · 49 marks

Questions

Question 1	*5 marks*	2
Question 2	*6 marks*	3
Question 3	*7 marks*	4
Question 4	*6 marks*	5
Question 5	*6 marks*	6
Question 6	*6 marks*	7
Question 7	*6 marks*	8
Question 8	*7 marks*	9

1. Prove by induction that

$$n^5 - n$$

is divisible by 5 for all positive integers n **[5]**

2. For each positive integer n, define

$$u_n = 5^{2n} - 2^n$$

Use proof by induction to show that 23 is a factor of u_n for all $n \in \mathbb{Z}^+$ **[6]**

3. Prove by induction that

$$9^n - 8n - 1$$

is divisible by 64 for all positive integers $n \in \mathbb{Z}^+$ **[7]**

4. Prove by mathematical induction that, for all non-negative integers n,

$$2^n + 2 \cdot 3^{2n} + 4^{2n+1}$$

is divisible by 7 **[6]**

5. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$7^n - 6n - 1$$

is divisible by 36 **[6]**

6. The function f is defined by

$$f(n) = 9^n - 8n + 63$$

Prove by induction that $f(n)$ is divisible by 64 for $n \geq 1$ **[6]**

7. Prove by mathematical induction that

$$7^{2n} + 8 \times 13^n - 9$$

is divisible by 36 for all positive integers n **[6]**

8. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$13^n + 3 \cdot 4^n + 11$$

is divisible by 12 **[7]**